

Synchronous memory will be faster than asynchronous memory in operations such as Windows startup, Photoshop image loading, gaming, uncompressed file copying. Toggle NAND memory is somewhere in between offering the cost benefits of the asynchronous memory with the performance benefits of synchronous memory.

Native garbage collection and TRIM support

It's a thumb rule that flash memory degrades over time. There is only a limited number of write cycles it can bear. A flash memory cell has large memory areas called 'blocks' which are made up of smaller memory chunks called 'pages'. Data is written in pages. So long as the pages in a flash memory are empty, there is no issue with adding data. But once the flash memory cell is full with data and you want to add more data, then it can get tricky, as data is not deleted page-wise, but block-wise. Whenever data needs to be written to a page, the entire block is written instead of just the particular page due to hardware limitations. So whatever data already exists needs to be stored in a temporary cache before it can be rearranged and then re-written onto the block. The maintenance of data involves two aspects, TRIM and garbage collection. A garbage collector is basically a dump for unwanted data. And data is termed/flagged unwanted by the controller is marked using the TRIM command. So with the use of both, the number of write cycles are effectively reduced, thus, increasing the SSD's life. TRIM is natively supported in Windows 7 and the latest Mac OS X iteration, but older versions of Windows such as XP and Vista do not, so make sure if you are on older systems, get a drive which supports native garbage collection. All new SSD support TRIM however, older gen SSDs which do not support TRIM are still available on the market so research well.

Size matters

Check out benchmarks and find one which matches your set of applications to figure out which SSD is best suited for your needs. For most users this will not make any difference, but we know our audience happens to consist of some ultra-geeks for whom every MB/s matters. There are various form factors available. The most common is 2.5-inch, followed

by the 1.8-inch SSD which in turn is suited for more compact configurations. Also, certain motherboards come with mSATA slots for making use of Intel's RST(Rapid Storage Technology). 60GB SSDs are best suited for this use, anything above is overkill.

Specs that don't

External casing

Many SSDs these days come with an SSD bracket, but an SSD-purchase decision should not hinge on this aspect. However, it's good to have.

Looks

SSDs will be placed inside your system, so external bright or metallic paints matter little. It's what inside that matters.

Recommendations

Corsair Force GS 240GB

This SSD sports toggle-mode NAND chip, which offers blazing speeds at the same time keeping the costs lower. We found the cost per GB to be lower than its Force GT drive and at a much better performance.

Kingston HyperX 120 GB

If you want a speedy SSD with quality components. Real life sequential tests easily crossed 200 MB/s whereas random operations were over 150 MB/s. Priced around ₹14,000 this drive makes sense if you are looking at for a drive to hold your OS as well as a couple of heavy applications and a game.

Crucial m4

One of the early entrants in the SSD market, the Crucial m4 series have only recently arrived in the Indian market. They offer great all-round performance.

Samsung 830 Series

Samsung has sent shivers down their competitor's spines with the 830 series. Also, since Samsung happens to be one of the major NAND chip makers globally, they have a home field advantage.

Intel 520 Series

If long term stability akin to enterprise offerings is something that you are interested in then the Intel 520 Series is the way to go. Recent price drops have made this a much more attractive purchase.

OCZ Agility

If massive amounts of storage at low cost is what you desire then you can opt for OCZ's agility series which provides the same while giving above average performance.

OCZ RevoDrive / Intel 910 Series

This is for people with deep pockets. PCI-e SSDs offer performance like no other, the sheer performance of PCI-e SSDs can only be matched if you have multiple 2.5-inch in RAID 0 off an enterprise RAID card.



When it comes to game consoles, you only have 3 devices to actually choose from, the XBOX360, PlayStation 3(PS3) and the Nintendo Wii; but the experience on each is worlds apart. How will you decide on which one to buy? Ideally, if you're a gamer, and you have the money and the time, we would say buy them all! They're all equally good and deciding between them is quite a difficult task. It's like trying to choose between a bike, car and SUV; they're entirely different beasts but essentially allow you to get from point A to B, but the experience and capabilities vary and in the end, it all boils down to your preference.

Stuff that matters Games

First, let's talk about the games. The Xbox 360 has some incredible exclusives such as Halo, Gears of War, Fable, etc. but then the PS3 has amazing exclusives such as GranTurismo, Metal Gear Solid, Uncharted, etc.; the Wii? Super Mario, Wii Fit and many more interesting titles. If you really have to pick one exclusively for gaming, it's best to go through the entire list of titles and see if anything specific appeals to you. Most of the popular games are available on both the Xbox and PS3,